

The Regolith Sinter Dome

A Mobile Microwave Sintering Rig for Building Pressure Shelters from Lunar Soil

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Classification: Speculative engineering, constrained to known physics

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Abstract

Everything sent to the Moon costs more per kilogram than almost anything on Earth is worth, which makes shipping building material there an absurdity — yet a lunar base needs thick, heavy walls to stop radiation and micrometeoroids. The material is already underfoot. This study describes the RS-1 'Hestia', a crawler that builds shelter out of the ground it stands on by microwave sintering: it lays regolith in courses and fuses each course into a solid ceramic mass in place, raising a vaulted dome without a single imported brick. The physics is unusually favourable, because lunar soil is not ordinary dust — micrometeorite bombardment has salted it with nanophase metallic iron that couples strongly with microwaves, so lunar regolith heats and fuses far more readily than terrestrial rock. We describe the rig, the sintering process, the structural logic of the dome, and — at length, because it is the part most often glossed over — the reason a sintered shell cannot simply be pressurised, and what must be done about it. This is among the most near-term concepts in either volume: the material is free, the process is laboratory-demonstrated, and the need is immediate.

1. Introduction: The Cheapest Material in the Solar System

A lunar habitat must be heavily shielded. Without an atmosphere or magnetic field, the surface is exposed to galactic cosmic rays, solar particle events, and a constant sleet of micrometeoroids, and the only broadly effective defence is mass — metres of it. Shipping that mass from Earth is out of the question; even at the most optimistic launch costs, a shielded habitat's worth of imported material would dwarf the cost of everything else in the programme. The economics of off-world construction therefore reduce to a single imperative: build with what is already there.

What is already there is regolith — a deep, loose blanket of pulverised rock and glass covering the entire lunar surface, produced over billions of years of impacts. It is abundant, free, and requires no mining in any meaningful sense; a scoop will do. The difficulty has always been binding it. Terrestrial construction binds aggregate with cement, which requires water and chemistry the Moon does not readily offer. The Sinter Dome's answer is to skip the binder entirely and fuse the regolith to itself with heat — turning loose dust into solid ceramic in place, using nothing but electricity and the soil underfoot.

2. Concept Description

The RS-1 'Hestia' is a crawler roughly the size of a small excavator, and it does three things in a repeating cycle. It gathers regolith with a simple scoop or blade; it lays that regolith as a shaped course along the growing wall of a dome, guided by a forming arm; and it passes a microwave applicator slowly over the fresh course, heating the material until the grains soften and fuse into a coherent solid. Then it advances a few centimetres and repeats. Over weeks, course by course, a vaulted shell rises out of the ground with no imported material and no human hands.

The geometry is deliberately ancient. A dome is the form that masonry has always used to span space without steel, because a properly shaped vault carries its loads in compression — and compression is exactly what fused regolith is good at, as with brick, stone, and concrete. Building in courses from the base upward, with each ring self-supporting as it closes, follows the same logic as a corbelled stone vault and needs no scaffolding or formwork, which is fortunate, since scaffolding would have to be shipped. The approach descends directly from terrestrial automated-construction research, in which large structures are built by machines depositing material layer by layer without moulds (Khoshnevis, 2004), adapted to a setting where the feedstock is free and the binder is heat.

3. Why Lunar Soil Sinters So Well

Sintering is the process of heating a powder until its particles bond into a solid mass without fully melting — the basis of ceramics and powder metallurgy for millennia. Any silicate dust can in principle be sintered, but lunar regolith turns out to be extraordinarily well suited to doing it with microwaves, for a reason peculiar to the Moon. Billions of years of micrometeorite impacts have vapour-deposited a coating of nanophase metallic iron onto the surfaces of the soil grains, and those iron particles couple strongly with microwave radiation.

The consequence, established experimentally, is that genuine lunar soil heats far more readily under microwaves than terrestrial simulants or ordinary rock — Taylor and Meek (2005) found that real lunar soil could be brought to melting in a modest microwave field with surprisingly little power, precisely because of this nanophase iron content, and set out the theory and practical implications for lunar construction. Subsequent work has characterised the high-temperature dielectric behaviour of lunar simulants to support process design (Allan et al., 2013). The practical upshot is that the energy budget for fusing regolith on the Moon is far more forgiving than an Earth-based intuition would suggest, and that microwave sintering — rather than solar furnaces, lasers, or chemical binders — is a strong candidate for the job. Microwaves also penetrate rather than merely warming a surface, so a course fuses through its depth instead of glazing on top. Control is everything, though: heat too little and the course stays friable, heat too much and it slumps and traps voids. The rig's real skill is holding the material in the narrow band between powder and puddle.

4. The Machine Itself

The crawler's demands are electrical and thermal rather than exotic. It needs a substantial, steady power supply — a solar array with storage for the long lunar night, or a connection to a surface grid such as the Planetary Power Loop described in Volume I — since sintering is fundamentally a matter of pushing energy into material. It needs a microwave generator and an applicator shaped to deliver that energy into a course of regolith at a controlled rate, and sensing to judge when a course is properly fused. It needs a scoop, a forming arm, and enough autonomy to lay courses accurately for weeks without supervision, since a dome built slightly wrong is a dome that falls down.

Two environmental problems dominate. Dust is the first: lunar regolith is abrasive, electrostatically clingy, and destructive to bearings and seals, and a machine that handles it all day must be built to survive that. Thermal cycling is the second — a fourteen-day day and an equally long, brutally cold night — which stresses both the rig and its work, since a freshly sintered mass cooling unevenly will crack. Controlled heating and cooling is central to the process, not incidental.

5. The Pressure Problem — and the Honest Answer

Here is the point that enthusiastic descriptions of sintered lunar domes usually skip, and it deserves to be stated plainly. A masonry dome works because gravity pushes its material together: the shell is in compression, and fused regolith, like stone and concrete, is strong in compression. But a pressurised habitat is the opposite of a masonry problem. Internal air pressure pushes outward, putting the shell into tension — and ceramics are weak in tension. Worse, lunar gravity is only about a sixth of Earth's, so the shell's own weight, which is what would normally hold it in compression, is correspondingly feeble. A bare sintered dome, pressurised to anything like a breathable atmosphere, would not hold; it would try to lift off its foundation and split.

There are two honest architectures that resolve this, and a serious design would use both. The first is to keep the pressure load off the ceramic entirely: the sintered dome serves as shielding and structure against the environment, while the pressure boundary is a separate flexible or metallic liner inside it, anchored independently. The dome protects; the liner contains. The second is ballast — burying the finished dome under a thick layer of loose regolith so that the weight of the overburden presses down on the shell and keeps it in net compression despite the internal pressure, which has the elegant side effect of adding exactly the radiation shielding the base needs anyway. Both approaches are consistent with the broader lunar-base engineering literature, which has long treated regolith mass as the primary shielding strategy and treated pressure containment as a separate problem (Benaroya, Bernold and Chua, 2002). The dome is a shield and a shell. It is not, by itself, a pressure vessel, and claiming otherwise is where these concepts usually go wrong.

Parameter	Phase 1 (Pads & berms)	Phase 3 (Habitat domes)
Product	landing pads, roads, walls	vaulted shells + overburden
Structural role	surface hardening	shielding shell (not pressure)
Pressure boundary	n/a	separate liner inside
Feedstock	regolith, unprocessed	regolith, unprocessed
Binder	none — microwave fusion	none — microwave fusion
Autonomy	supervised passes	weeks of unattended courses

Table 1. Representative parameters for the staged build-out.

6. Operations Concept

Phase one is the unglamorous work that pays for itself immediately: sintering landing pads and roadways. Rocket exhaust striking loose lunar dust throws it outward at high speed, sandblasting nearby hardware — a serious and well-documented hazard — so a fused, hardened pad is among the first pieces of infrastructure any lunar base actually needs, and it requires no structural sophistication at all. Phase two builds berms, retaining walls, and radiation shielding around existing modules, still in pure compression and still structurally simple. Phase three raises full vaulted domes over prepared sites, into which a separate pressure liner is fitted and over which loose regolith is banked as ballast and shielding. The progression is deliberately conservative: each phase delivers something useful and each earns the next.

7. Honest Difficulties

Beyond the pressure question, several obstacles are real. Power is the largest: fusing significant volumes of material demands sustained kilowatts, and surviving the fourteen-day lunar night without stopping work requires either large storage or an external supply. Process control is the second — the window between under-sintered crumble and over-melted slump is narrow, and regolith composition varies from place to place, so the rig must sense and adapt rather than follow a fixed recipe (Allan et al., 2013). Dust degradation of mechanisms is chronic and unsolved in the general case. Cracking from thermal cycling threatens both fresh and finished work. And the resulting material, while strong in compression, is a brittle ceramic with the failure behaviour ceramics have — it does not bend before it breaks, which constrains how it can safely be used. None of these is a barrier of physics; the fusion itself is demonstrated (Taylor and Meek, 2005).

8. Pathway to Reality

This concept sits close to the present. Microwave sintering of lunar soil and simulants has been studied experimentally for two decades (Taylor and Meek, 2005; Allan et al., 2013); automated layer-by-layer construction is a mature terrestrial technology (Khoshnevis, 2004); regolith-based construction and shielding are central to serious lunar-base engineering studies (Benaroya, Bernold and Chua, 2002); and space agencies and private programmes are actively developing in-situ construction demonstrations for the coming lunar missions. The credible near-term step is a small sintering rig producing a landing pad — a genuinely valuable deliverable that needs no structural breakthrough — with domes following once the process, the power, and the autonomy have been proven at that lower stake.

9. Conclusion

Every civilisation before ours built its first shelters from whatever the ground gave it, and there is no reason for the next one to be different. The Regolith Sinter Dome takes the cheapest material in the solar system — the dust that is already everywhere, that we would otherwise merely curse for clogging our machines — and turns it into pavement, walls, and vaults using nothing but electricity and a peculiarity of lunar chemistry that makes that dust unusually eager to fuse. It will not, on its own, hold air; that honest limitation shapes the whole design. But it will hold back the radiation, the micrometeoroids, and the blasted dust of every landing, and it will do so without a single kilogram lifted from Earth. The first true lunar building will not be delivered. It will be baked in place.

References

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- Allan, S.M., Merritt, B.J., Griffin, B.F., Hintze, P.E. and Shulman, H.S. (2013) 'High-temperature microwave dielectric properties and processing of JSC-1AC lunar simulant', *Journal of Aerospace Engineering*, 26(4), pp. 874-881.
- Benaroya, H., Bernold, L. and Chua, K.M. (2002) 'Engineering, design and construction of lunar bases', *Journal of Aerospace Engineering*, 15(2), pp. 33-45.
- Khoshnevis, B. (2004) 'Automated construction by contour crafting - related robotics and information technologies', *Automation in Construction*, 13(1), pp. 5-19.
- Taylor, L.A. and Meek, T.T. (2005) 'Microwave sintering of lunar soil: properties, theory, and practice', *Journal of Aerospace Engineering*, 18(3), pp. 188-196.